

EXPERIENCE

Graduate Course Developer / Head TA
University of Pennsylvania

May 2024 - Present

Designed and instructed a graduate-level course on scalable IoT systems for industrial applications, covering CI/CD pipelines, fleet management, RTOS development (**Zephyr**, **FreeRTOS**), wireless protocols (**LoRa**, **BLE**, **Wi-Fi**, **Cellular**), and PCB design. Modernized the course's technical infrastructure by integrating the latest wireless microcontrollers, enabling broader application scenarios and simplifying the design and development of embedded software.

Research Assistant
Penn Computational Intelligence Lab (PennCIL)

May 2024 - Aug 2024

Prototyped an analytical benchmarking tool to support robotic hardware-software co-design by modeling trade-offs between minimal compute architectures and the effectiveness of various control algorithms—derisking the time and resource intensive physical testing when designing custom robotic hardware.

Research Assistant
Mobile Manipulator Project, Nadia Figueroa Lab

Sep 2023 - May 2024

Upgraded the onboard Jetson computer of a SCOUT II rover by reconfiguring the full software stack, transitioning the system to **ROS 2** and writing custom scripts to streamline data collection and operational workflows

Undergraduate Researcher
ULoc, Wireless Communication, Sensing and Networking Group (WCSNG)

Apr 2023 - Oct 2023

Tuned and calibrated a cluster of IMU and UWB sensors to enable accurate VR body orientation and position tracking. Conducted comparative analysis of various filtering and state estimation models to evaluate and benchmark the lab's sensor fusion pipeline against other IMU-based VR tracking systems.

Electronics Executive
Triton Robotics (TR)

Jan 2020 - Jul 2023

Founded and built the electronics team to develop hardware solutions that enhanced robotic performance and maintainability. Primary projects included prototyping a super-capacitor augmented power PCB and designing a custom STM32 Nucleo shield that restructured the electrical architecture for improved maintainability, bandwidth, and power protection.

SDE Intern
Amazon Inc

Jun 2022 - Sep 2022

Redesigned a previously manual anomaly review system into scalable tools, pipelines, and APIs, streamlining data audits and enabling global teams to efficiently monitor and resolve ML pipeline issues for the Amazon website.

EDUCATION

University of Pennsylvania (UPenn)

M.S. in Robotics

Aug 2023 - May 2025

GPA: 3.87

University of California, San Diego (UCSD)

B.S. in Computer Engineering

Sep 2019 - Jun 2023

GPA: 3.925

Inducted IEEE Eta Kappa Nu
March 2022

SKILLS

Control Algorithms, Trajectory Optim. Embed. System, ROS, PCB design, CAD (Solidworks), API/Backend Dev., C/C++, Python, Git, Docker, Sensors